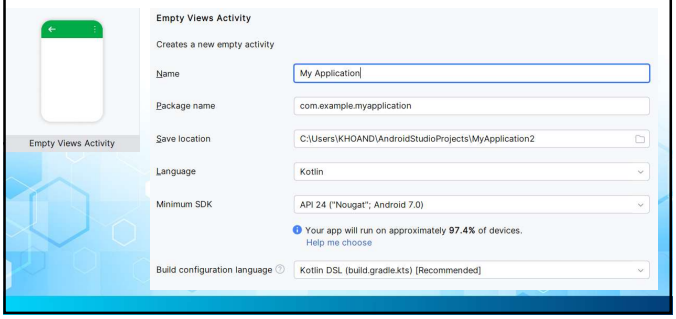
COS30017. SOFTWARE DEVELOPMENT FOR MOBILE DEVICES

Dr. Nguyen Dang Khoa
Email: kdnguyen@swin.edu.au
Phone: 0974.844.618

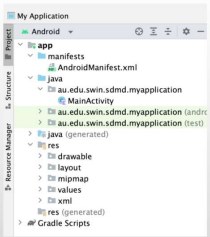
Week 2 - Kotlin



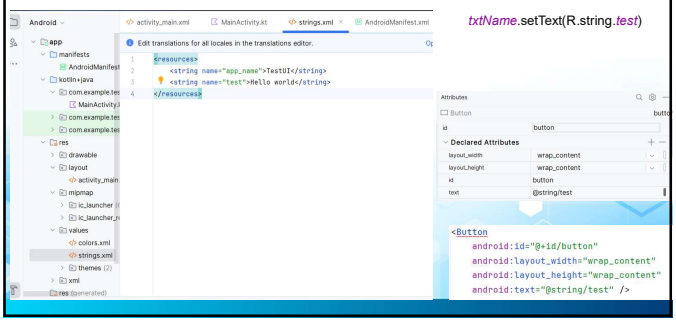
Week 3 – App building blocks

Android Project Structure (convention)

- Source code (**java**)
- Resources (**res**)
  - Images (**drawable**)
  - Layout of app (**layout**)
  - Launcher icons (**mipmap**)
  - Constants/Strings (**values**)



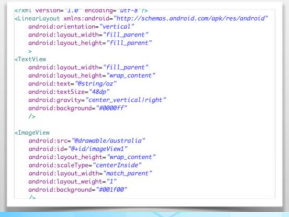
Resources



Week 3 – App building blocks

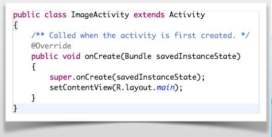
Presentation

- Layout definition (activity_main.xml)



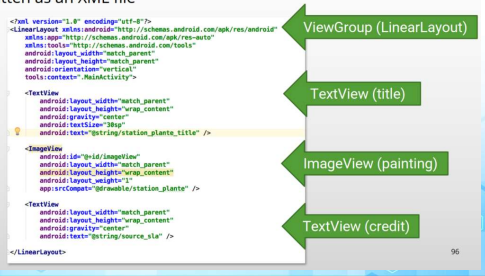
Functionality

- Activity class (MainActivity.java):
  - Event handling, I/O …

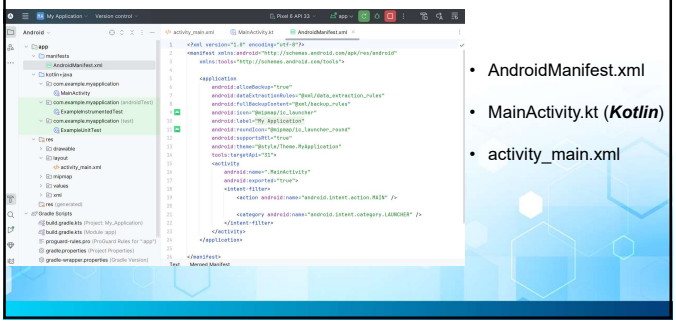


Week 3 – App building blocks

- Layout can be written as an XML file

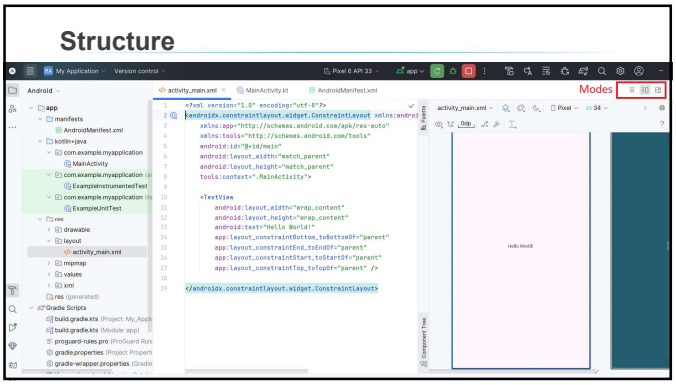


Structure

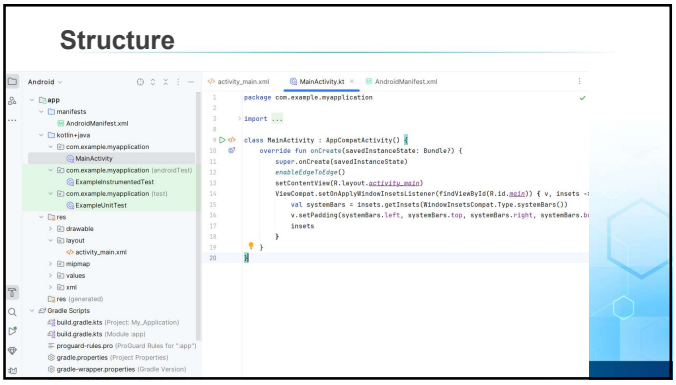


- AndroidManifest.xml
- MainActivity.kt (***Kotlin***)
- activity_main.xml

Structure

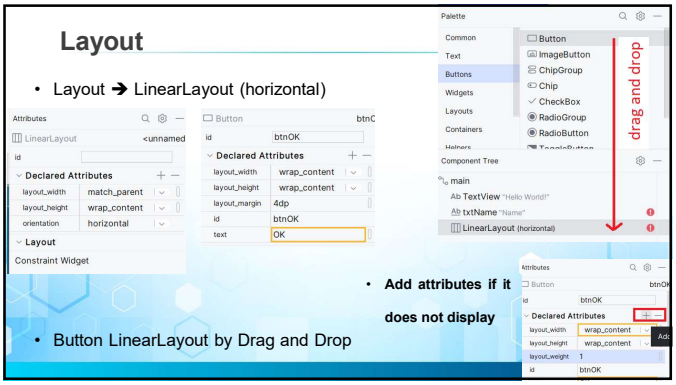


Structure



Layout

- Layout ➔ LinearLayout (horizontal)

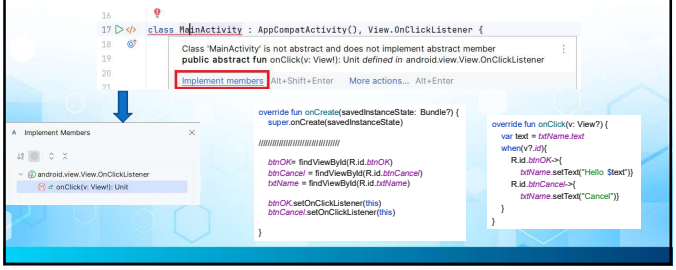


- Button LinearLayout by Drag and Drop
- **Add attributes if it does not display**

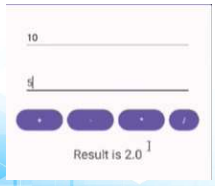
Click event

```
lateinit var btnOK:Button;
lateinit var btnCancel:Button;
lateinit var  txtName:EditText;
```

```
class MainActivity : AppCompatActivity(), View.OnClickListener {
```

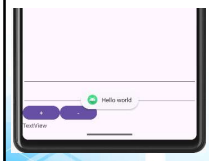


Experimentals




```
var text = txtName.text.toString().toDouble()
```

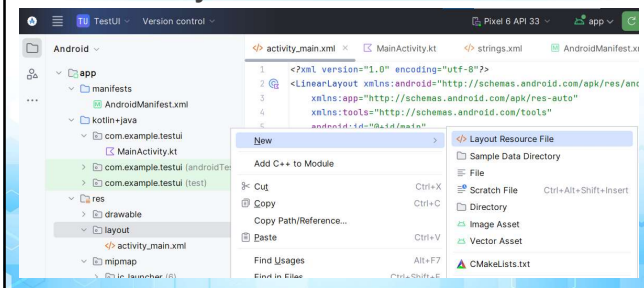
Toast



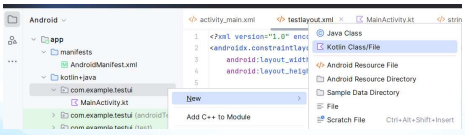

```
when(v?.id){
    R.id.btnOK->{
        val c1 = Car("Ford", "Mustang", 1969)
        txtName.setText(c1.GetBrand())

        Toast.makeText(this, "Hello world", Toast.LENGTH_LONG).show()
    }
```

New activity



Right click layout ➔ New ➔ Layout Resource File ➔ ***testlayout.xml***



Right click kt code ➔ New ➔ Kotlin class/File
➔ ***testActivity***

XML file

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:id="@+id/main"
    tools:context=".testActivity" >

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click me" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

xmlns:tools="http://schemas.android.com/tools"

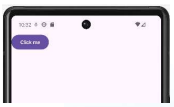
android:id="@+id/main"

tools:context=".testActivity" >

Kotlin file

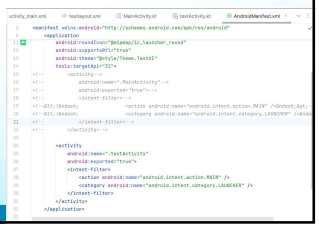
```kotlin
import android.os.Bundle
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat

class testActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        enableEdgeToEdge()
        setContentView(R.layout.testlayout)
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v, insets ->
            val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars())
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
            insets
        }
    }
}
```



AndroidManifest.xml ➔ new activity

```xml
<activity
    android:name=".testActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
```



Change activity

```
<activity
    android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>

<activity android:name=".testActivity">

</activity>
```

```
val intent = Intent(this, testActivity::class.java)
startActivity(intent)
```

Practical

- Using the ImageView to display one image
- Using RES

class LoginActivity

```
lateinit var btnLogin: Button
lateinit var txtUser: EditText
lateinit var txtPass: EditText

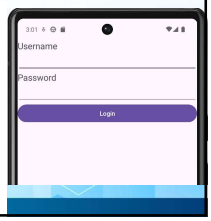

btnLogin= findViewById(R.id.btnLogin)
txtUser = findViewById(R.id.txtUser)
txtPass = findViewById(R.id.txtPass)
btnLogin.setOnClickListener(this)


override fun onClick(v: View?) {
    when (v?.id) {
        R.id.btnLogin -> {
            if ((txtUser.text.toString()=="KhoaND") && (txtPass.text.toString()=="123"))
            {
                //Toast.makeText(this, "Correct", Toast.LENGTH_LONG).show()
                val intent = Intent(this, AdminActivity::class.java)
                startActivity(intent)
            }
            else
                Toast.makeText(this, "Not correct", Toast.LENGTH_LONG).show()
        }
    }
}
```

AndroidManifest.xml

```
<activity
    android:name=".LoginActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>

<activity android:name=".AdminActivity"></activity>
```

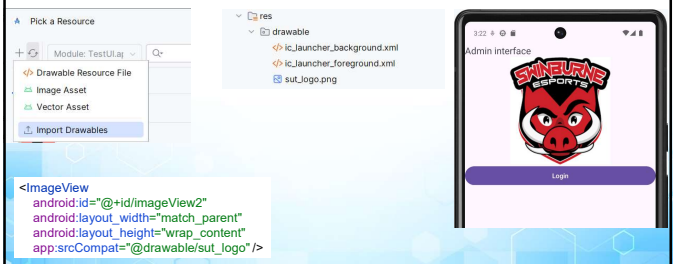


AndroidManifest.xml

```
<activity
    android:name=".LoginActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>

<activity android:name=".AdminActivity"></activity>
```

ImageView




```
<ImageView
    android:id="@+id/imageView2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    app:srcCompat="@drawable/sut_logo" />
```

Content

- Loop, toList, filter, until
- If, else, when
- Structure of directory: res, layout, drawable, mipmap, values...
- Layout present in xml (Constraint, Linear...)
- ImageView, Button, TextView...